Session 1
Key concepts for automated classification and introduction to bag-of-words approaches

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LISS2117 · *Quantitative methods for text classification and topic detection*

Programme

- Introduction
  - Who?
  - What?
  - How?
- Text-as-data
- Bag-of-words representations
- Basic approaches to classification
- Validation of results

Who

Who: Dr. Michele Scotto di Vettimo

- **About me**

  Political scientist, PhD at King's College London (2022)

  Postdoc at the Centre for Computational Social Sciences of the University of Exeter

  Currently Research Associate in the Department of Political Economy at King's College London

- **Get in touch**

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- my research: **European Union | public opinion | policy responsiveness | non-majoritarian institutions**

What

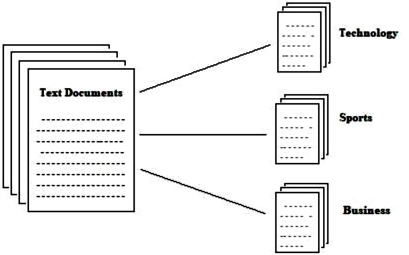
What: automated classification of texts

We will:

focus on a specific task among those that can be performed with text-as-data approaches: **text classification**;

highlight its conceptual flexibility and proximity with other tasks;

cover various ways in which it can be implemented.



How

How: course format

The course is designed to be useful for students with different levels of familiarity with text-as-data approaches to classification (e.g., should be useful both to those who are considering using these methods but still know little about them, and to those that are already more experienced but want to learn more about a specific technique).

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Required readings cover essential aspects of what will be discussed in class. Hence, you should read them before our classes. Optional readings give you additional food-for-thought and examples of applications of specific methodologies.

How: course structure

- **Outline**:

  Session 1: introduction, key concepts and basic approaches (May 7, 2025)

  Session 2: topics models and machine learning algorithms (May 14, 2025)

  Session 3: word-embeddings and large language models (May 21, 2025)

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- **Practicalities**:
  - Sessions runs from 10am to 4pm
  - There will be short breaks as appropriate, as well as a longer break around 1pm
  - Slides will be shared the day before each session
  - All materials can be accessed on the KEATS page of the course, or at this page
  - A full reading list with additional readings is provided here
  - We will use `R` and Python coding languages. Please refer to this document for guidance on setting up the software.

Text-as-data

Text-as-data: an overview

- Text as Text *versus* Text as Data
  - "Ironically, generating insight from text as data is only possible once we have destroyed our ability to make sense of the texts directly. To make it useful as data, we had to obliterate the structure of the original text [...]." [Benoit et al., 2020]

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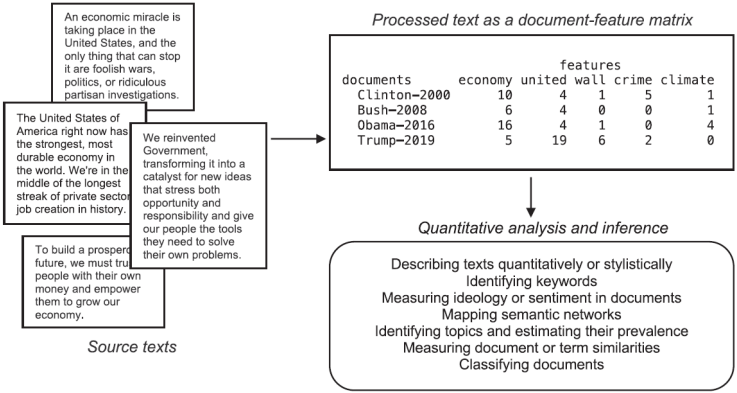

Text-as-data: an overview

- Text as Data approaches cover a wide variety of methods and tasks

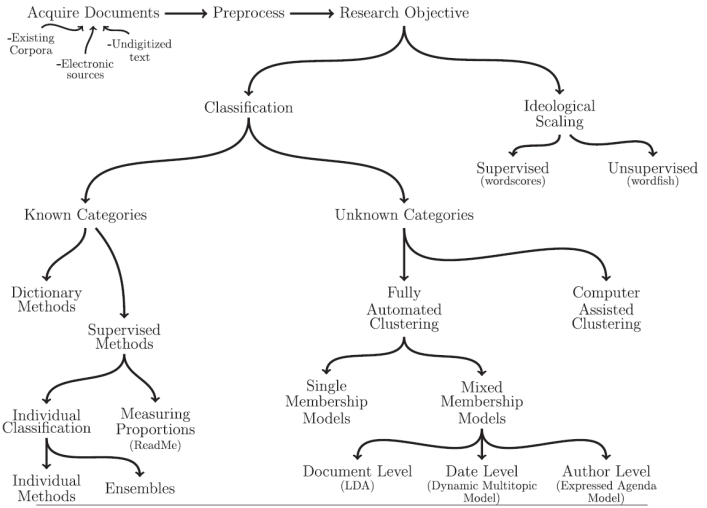

Fig. 1 An overview of text as data methods.

[Grimmer and Stewart, 2013]

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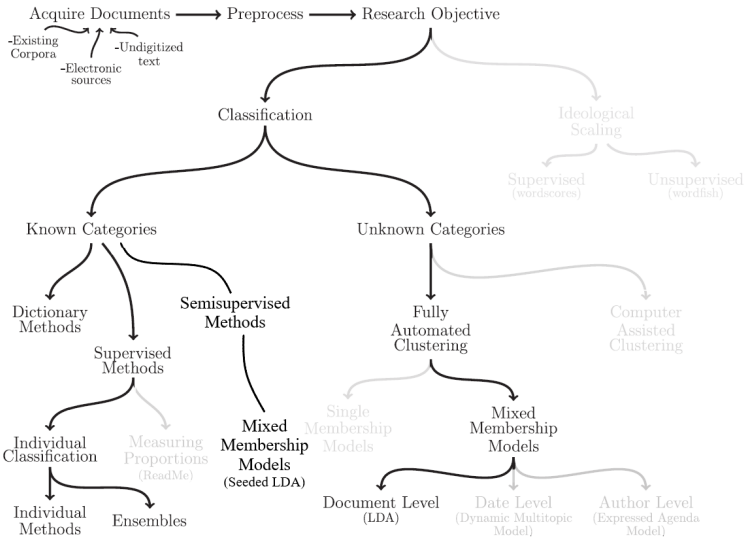

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5. Validation of the results
   - How can we convince someone that our selected method works?

Bag-of-words representations

Bag-of-words representations: key concepts

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Key to this representation is the "**document-feature matrix**" (abbreviated **dfm**; sometimes called also "document-term matrix"):

	about	bird	heard	is	the	word	you
About the bird, the bird, bird bird bird	1	5	0	0	2	0	0
You heard about the bird	1	1	1	0	1	0	1
The bird is the word	0	1	0	1	2	1	0

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- "White_House" is a *n*-gram, an ordered set of *n* words
- *n*-grams count as single tokens (so sentence *a*. has 8 tokens, whereas sentence *b*. has 7), and they can be introduced in the tokenisation every time we need to take into account multi-word expressions or order of words in the dfm.

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Note: Pre-processing is useful and generally advised when working with bag-of-words approaches. However, that is not the case when working with embeddings representation (in fact, it can harm performance)

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These steps help reducing the size of the final document-feature matrix, thus making subsequent analysis easier and faster.

However, you should always think carefully about the appropriateness (or even just the ordering!) of these pre-processing steps in the context of your research focus.

Rethinking this default procedure and tailoring it to your needs and to the specificities of the selected text-as-data method is key to avoid a poor use of the available data [Grimmer et al., 2022, 57-59]

We will be mostly relying on the `quanteda` package:

- `corpus()`
- `tokens()`
- `dfm()`
- `tokens_*()`, `dfm_*()`

When lost, cry for `help()`! Like this: `help(corpus)`

Basic approaches to classification

Basic approaches: keyword counting and dictionary methods

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Both approaches stand on the assumption that human-defined rules at the token level can be used to determine labels (assignment into a category) at the text level

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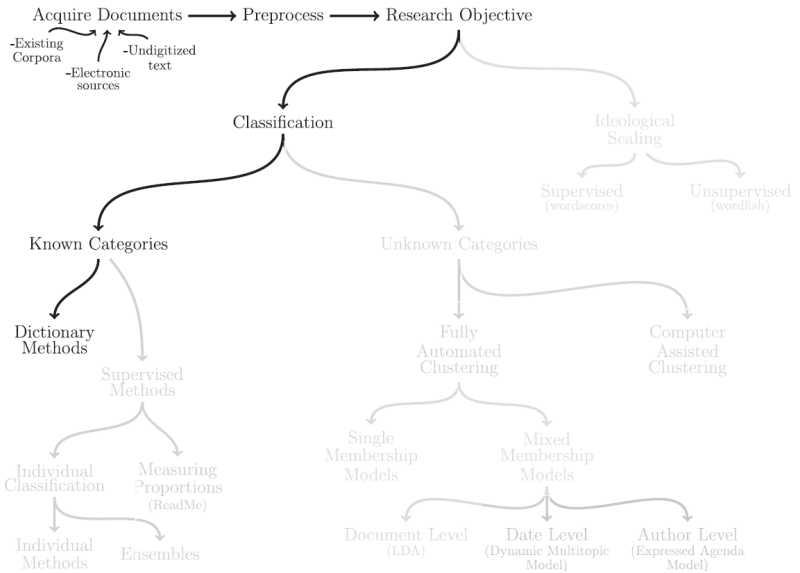

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- Keyword counting
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You can find examples of applications of keyword counting and dictionary methods among the readings under "Dictionary methods and semantic scaling" on this page.

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- Cons
  - Dictionary keywords might be very context dependent
  - Sometimes the concept we are interested is nuanced, and keywords would not capture it satisfactorily
  - Keywords can be ambiguous
  - Exhaustive dictionaries can become very long
  - Need to anticipate evolution of language (or be updated regularly)

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- As a compromise, one can start from a validated dictionary and make additions or deletions so as to adapt it to the research needs

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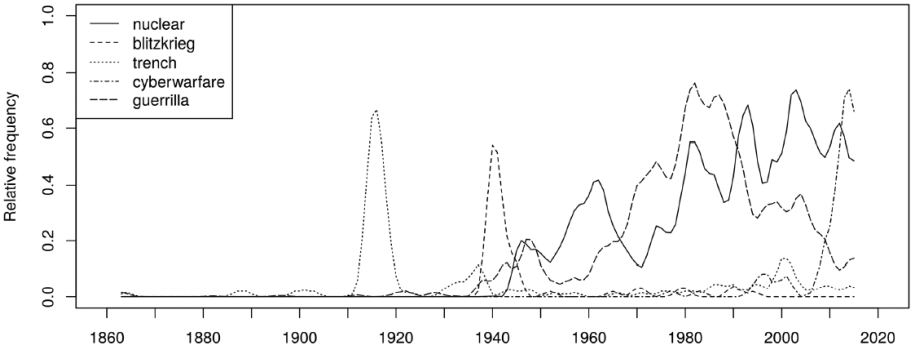

Figure 6. Frequency of keywords for international ideologies and military strategies.

[Trubowitz and Watanabe, 2021]

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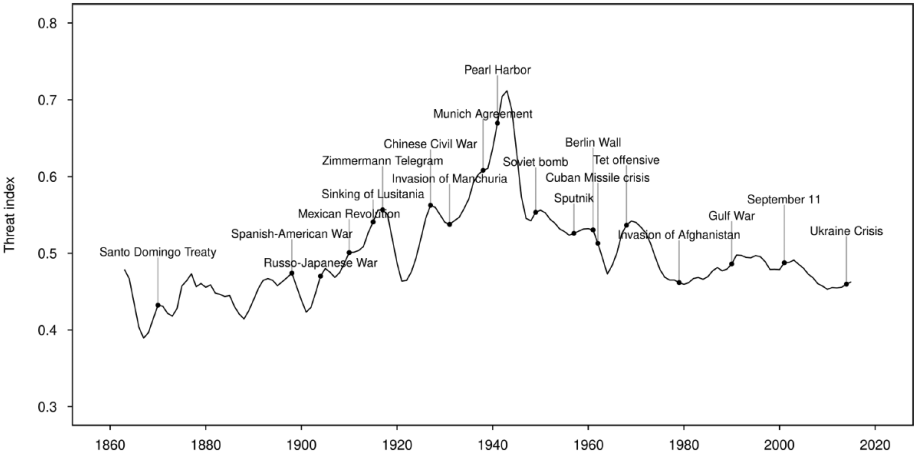

Figure 7. Baseline GTI for United States, 1861–2017. (Based on full sample of 225 countries and kernel smoothed by ±1 year.)

[Trubowitz and Watanabe, 2021]

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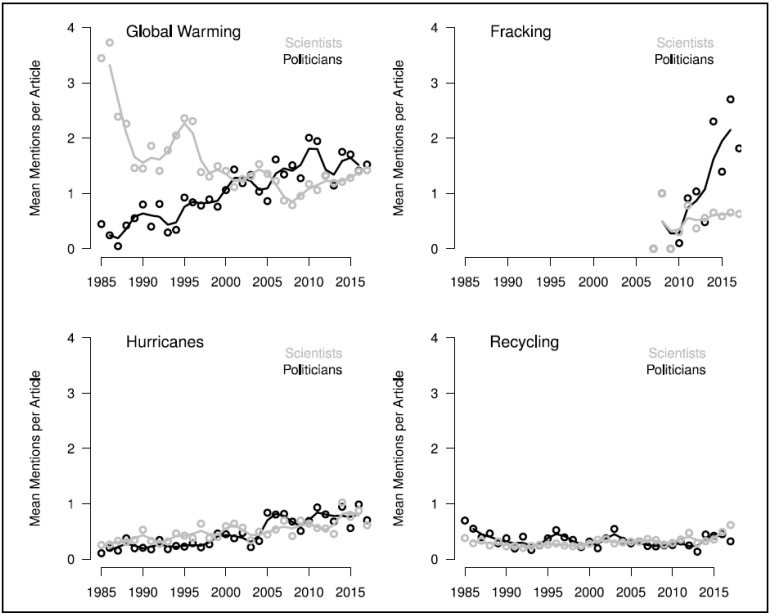

Figure 1. Politicization in news coverage of environmental issues.

[Chinn et al., 2020]

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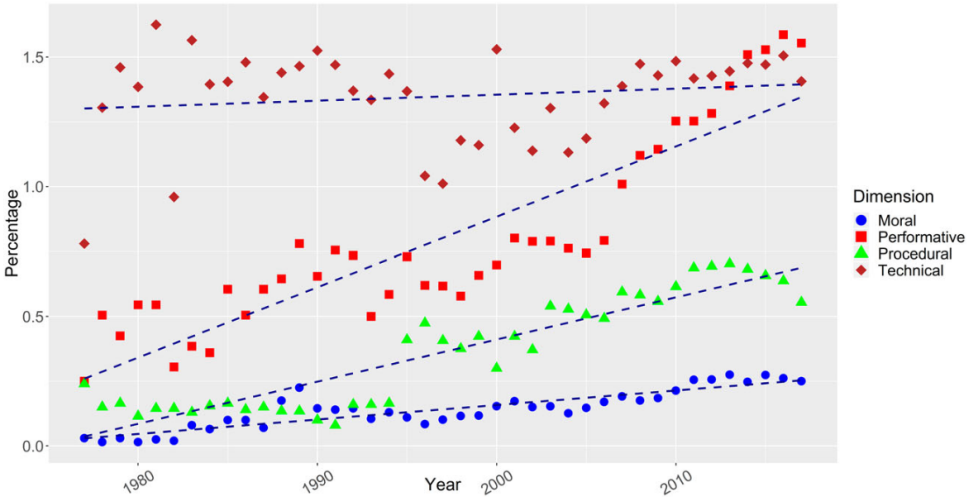

Figure 1. Dimension trends.

[Busuioc and Rimkutė, 2020]

We will practice with:

- `kwic()`
- `tokens_select()`
- `tokens_lookup()`
- `dictionary()`

We will create a customised `R` function with `function()`

When lost, cry for `help()`!

Validation of results

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"***Validate, validate, validate!***" [Grimmer and Stewart, 2013]

For text-as-data methods, validation essentially means showing that the measures produced using our method actually work, and enable us to make valid social science inferences [Grimmer et al., 2022].

Validity assessments take different forms (and this is true not just for text-as-data methods [Adcock and Collier, 2001]):

- Does our measure pass the inspection of a subject expert or conform with established general knowledge? (*face validity*)
- Is our measure appropriate for the way our concept of interest is theorised and understood? (*content validity or fidelity*)
- Is our measure behaving as we would expect with regard to established set of patterns or obvious hypotheses? (*hypothesis or convergent validity*)

Validation of results

More practical tips:

- Use the method to replicate a task on data for which we know the correct classification (kind of unavoidable in a supervised context)
- Read texts assigned to a particular category and assess their coherence (useful in semi or unsupervised settings)
- Examine other variables or features associated with a specific category

Validation of results

Dictionaries need explicit validation [Grimmer et al., 2022]

- If "gold-standard" annotation is available for part of the texts, then assess how good is the dictionary at replicating it
- Do your own coding: produce hand labels for a sample of texts and compare with dictionary classification (but do not cheat! use a separate validation data)
- Check if your categories correlate with other observable variables measured independently
- We will cover validation more extensively in the next session

No new commands or functions, just more practice with the ones already introduced.

When lost, cry for `help()`!

Unsupervised classification

Unsupervised, semisupervised, and supervised methods

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  Supervised methods: trained on labelled texts to learn relations between tokens and labels and classify texts (labelled training data required)

Recap

- Text-as-data methods requires transforming raw texts into numerical data for analysis and inference
- There are various ways of representing texts numerically, and bag-of-words representations are the most basic
- Automated text classification is a type of text-as-data tasks that maps tokens to classes or labels
- Keyword counting and dictionary methods are two simple approaches to detect topics or assign documents into classes, and both rely on bag-of-words representations
- They use keywords identified by the analyst to map documents to labels (simple and intutive, but also highly context-dependent and lack nuance)
- Even if basic and intuitive, like all other methodologies they require careful validation

Next session:

Topics models and machine-learning algorithms for classification

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